

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1708

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.(BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour

Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I JASMITHA R (192524295) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

JASMITHA R

Annexure A

Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that: Constraints:

- i. D1 cannot work Night shift**
- ii. D2 must work before D3 (Morning < Afternoon < Night)**
- iii. Only one doctor per shift**
- iv. D3 cannot work Morning**
- v. All doctors must be assigned exactly one shift**

Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

1. Constraint Satisfaction Problem (CSP) using Backtracking Search (10 Marks)

ANSWER:

Problem Statement: A hospital needs to assign **3 doctors (D1, D2, D3)** to **3 shifts (Morning, Afternoon, Night)** while satisfying the following constraints.

Doctors

- D1
- D2
- D3

Shifts

- Morning (M)
- Afternoon (A)
- Night (N)

Constraints

- 1. D1 cannot work Night shift.**
- 2. D2 must work before D3 (Morning < Afternoon < Night).**
- 3. Only one doctor per shift.**
- 4. D3 cannot work Morning shift.**

5. Each doctor is assigned exactly one shift.

Initial State

No doctor is assigned.

Doctor	Shift
D1	-
D2	-
D3	-

Backtracking Search Steps

Step 1: Assign D1

Possible shifts for D1:

- Morning YES
- Afternoon YES
- Night NO (Constraint 1)

Assign:

D1 → Morning

Current Assignment:

Doctor	Shift
D1	Morning
D2	-
D3	-

Step 2: Assign D2

Available shifts:

- Afternoon
- Night

Try:

D2 → Afternoon

Current Assignment:

Doctor	Shift
D1	Morning
D2	Afternoon
D3	-

Step 3: Assign D3

Available shift:

- Night

Check Constraints:

- D3 cannot work Morning
- D2 works before D3 (Afternoon < Night)
- Only one doctor per shift

Assign:

D3 → Night

Constraint Checking

Constraint	Status
D1 not Night	✓ Satisfied
D2 before D3	✓ Afternoon before Night
One doctor per shift	✓ Satisfied
D3 not Morning	✓ Satisfied
Every doctor assigned once	✓ Satisfied

Backtracking Search Tree

Start

|

| — D1 → Morning ✓

| |

| | — D2 → Morning ✗ (Already occupied)

| |

| | — D2 → Afternoon ✓

| | |

| | | — D3 → Night ✓ Goal

| |

| | — D2 → Night ✗

| |

| | — D3 → Afternoon ✗

| (D2 must be before D3)

|

|— D1 → Afternoon

| (Alternative branch)

|

|— D1 → Night **X**

(Constraint violated)

Final Valid Schedule

Doctor	Assigned Shift
D1	Morning
D2	Afternoon
D3	Night

Algorithm (Backtracking Pseudocode)

BEGIN

Assign shifts to doctors one by one.

For each doctor:

 Choose an available shift.

 Check all constraints.

 If constraints are satisfied:

 Assign the shift.

 Move to the next doctor.

 Else:

 Backtrack and try another shift.

Repeat until all doctors are assigned.

Display the final schedule.

END

Conclusion

Using the **Backtracking Search** algorithm, the constraints are checked after every assignment. If a constraint is violated, the algorithm backtracks and tries another assignment. The final valid schedule is:

- **D1 → Morning**
- **D2 → Afternoon**
- **D3 → Night**

This assignment satisfies all the given constraints and is the required solution.

2.A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

Constraints:

- Movement allowed: Up, Down, Left, Right**
- Cost of each move = 1**
- Cannot pass through obstacles**
- Use Manhattan Distance heuristic**

Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

ANSWER:

Constraints:

- Up, Down, Left, Right movement
- Cost per move = 1
- Cannot cross obstacles
- Manhattan Distance (used in informed search like A*, not BFS)

BFS Node Expansion

Step	Expanded Node
1	S (0,0)
2	(1,0)
3	(0,1)
4	(2,0)
5	(2,1)
6	(2,2)
7	(3,2)
8	(3,3)
9	(3,4)
10	G (4,4)

Optimal Path

$S \rightarrow (1,0) \rightarrow (2,0) \rightarrow (2,1) \rightarrow (2,2) \rightarrow (3,2) \rightarrow (3,3) \rightarrow (3,4) \rightarrow G$

Total Cost

- Number of moves = 8
- Cost per move = 1
- **Total Cost = 8**

BFS Algorithm

1. Insert Start node into Queue.
2. Remove front node.
3. Expand all valid neighbors.
4. Add unvisited neighbors to Queue.
5. Repeat until Goal is reached.
6. Return shortest path.

Conclusion

Optimal Path: $S \rightarrow (1,0) \rightarrow (2,0) \rightarrow (2,1) \rightarrow (2,2) \rightarrow (3,2) \rightarrow (3,3) \rightarrow (3,4) \rightarrow G$

Minimum Cost = 8

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

- a) Analyze all the given constraints and explain how they affect the robot's decision-making process.
- b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.
- c) Demonstrate how the robot applies AI concepts such as:
 - Intelligent agents
 - Rationality
 - Environment type

Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

ANSWER:

Problem Statement

An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot starts at S and must reach the survivor at G. The environment is partially observable, meaning the robot can only sense nearby cells. It must avoid obstacles, minimize travel cost, and update its knowledge while exploring.

a) Analyze the Constraints and Their Effect on Decision-Making

1. Movement Constraint

- The robot can move only **Up, Down, Left, and Right**.
- Diagonal movement is not allowed.
- The robot must choose the best direction at every step.

2. Obstacles

- Some rooms are blocked.
- The robot cannot move through these cells.
- It must search for an alternate path.

3. Limited Sensing

- The robot can observe only adjacent cells.
- It does not know the complete map initially.
- It discovers the environment while moving.

4. Movement Cost

- Normal movement cost = **1**
- Entering a risky zone costs **3** ($1 + 2$)
- The robot prefers safer routes with lower total cost.

5. Dynamic Knowledge

- The robot continuously updates its map after sensing new cells.
- Future decisions depend on newly discovered information.

6. Goal

- Reach the survivor safely.
- Minimize the total travel cost.
- Avoid dangerous and blocked paths.

b) Solution Approach using Online Search Agent and Uniform Cost Search

Why Uniform Cost Search (UCS)

Uniform Cost Search expands the path with the lowest cumulative cost first.

Since risky areas have higher costs, UCS naturally chooses safer and cheaper paths.

Algorithm Steps

Step 1: Start from node S.

Step 2: Sense all adjacent cells.

Step 3: Update the internal map.

Step 4: Calculate cumulative cost for each possible move.

Step 5: Expand the node with the smallest cost.

Step 6: Ignore blocked cells.

Step 7: Continue exploring until **G** is reached.

Step 8: Return the least-cost path.

Pseudo Code

BEGIN

Place Robot at Start (S)

WHILE Goal not reached

 Sense adjacent cells

 Update environment map

 Expand least-cost node

 Ignore blocked cells

 Move to safest path

END WHILE

Return optimal path

END

c) AI Concepts Applied

1. Intelligent Agent

The rescue robot is an intelligent agent because it:

- Senses the environment.
- Makes decisions independently.
- Updates its knowledge.
- Performs actions automatically.

2. Rationality

A rational agent always chooses the action that gives the best outcome.

The robot:

- Chooses the safest path.
- Avoids risky areas whenever possible.
- Minimizes total travel cost.
- Successfully reaches the survivor.

3. Environment Type

Property	Description
Partially Observable	Robot sees only nearby cells.
Dynamic	Environment information changes while exploring.
Sequential	Each move affects future decisions.
Discrete	Robot moves between grid cells.
Single Agent	Only one rescue robot operates.

Why Online Search Agent + Uniform Cost Search

- Works in unknown environments.
- Continuously updates knowledge.
- Finds the minimum-cost path.
- Avoids obstacles and risky zones.
- Suitable for disaster rescue operations where complete information is unavailable.

Advantages

- Finds the optimal path.
- Handles varying movement costs.
- Adapts to newly discovered obstacles.
- Improves safety and rescue efficiency.
- Reduces total travel cost.

Conclusion

The Online Search Agent combined with Uniform Cost Search (UCS) is the most suitable approach for the autonomous rescue robot. It allows the robot to sense nearby cells, update its knowledge dynamically, avoid obstacles, minimize travel cost, and safely reach the survivor. This approach demonstrates key AI concepts such as intelligent agents, rational decision-making, and search in a partially observable environment, making it ideal for disaster rescue applications.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided